

A Minimal UV Deformation Forced by the Axial Obstruction (Not a Selected Ultraviolet Completion)

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15 August 2026 — Version 1

Abstract

Paper 11 refused to invent a CFT, a string compactification, or an asymptotically safe fixed point to fill the ultraviolet line of the Emergent Gravity Program. That refusal stands. SuperGrok does not repeal it.

Paper 14 nevertheless proved an obstruction: Einstein–Cartan Hehl–Datta is a contact, while CFT axial Fisher is nonlocal. Any local bulk that matches the latter *and* reduces to the New Standard Model in the infrared must therefore *not* be Einstein–Cartan at all scales. The unique quadratic deformation that does this, with the M9.1 coefficient locked in the infrared, is a massive axial field whose on-shell effective kernel is

$$r(k) = \frac{3/16}{1 + k^2/M^2}.$$

Tree-level matching is derived and machine-checked. The position-space kernel is the four-dimensional Yukawa function, not a delta. The massless limit kills the contact.

This is a deformation, not a selection. It does not produce the Standard Model, it does not renormalize the metric, and it does not name a holographic pair. Final Status “UV completion” remains [O] for that selected-pair question. The axial deformation itself is [P] as tree-level effective-field-theory algebra and [C] as the physical identification of S_μ with axial torsion.

No [O] is moved to [P] for Q4a.

Standard of proof

“A UV that works” is split before any Lagrangian is written.

1. **Q4a (selected completion).** A microscopic theory whose low-energy limit *is* the New Standard Model, including the installed gauge group, representations, Yukawas, and a ultraviolet-complete metric sector. Inventing one to fill a section is a fabrication [?]. Still [O].
2. **Q4b (forced deformation).** The minimal local bulk extension that (i) reproduces $\mathcal{L}_{\text{HD}} = -(3\kappa/16) J_5 \cdot J_5$ as $M \rightarrow \infty$, and (ii) has a nonlocal axial quadratic form at finite M , so that Paper 14 Theorem obs is not a contradiction. This is an inverse problem with a unique *quadratic* solution once the infrared lock is imposed. That uniqueness is inside the ansatz, not in the space of all quantum gravities.

Numerical agreement is not proof. The algebra below is the proof of Q4b at tree level. A script is a check.

1 Why Einstein–Cartan cannot be its own UV

The New Standard Model after torsion elimination is [?, ?]

$$\mathcal{L}_{\text{IR}} = \frac{1}{2\kappa} \mathring{R} - \Lambda + \mathcal{L}_{\text{SM}} - \frac{3\kappa}{16} J_5^\mu J_{5\mu}, \quad \kappa = 8\pi G. \quad (1)$$

The last term is ultralocal. In any unitary CFT in $d \geq 2$,

$$\langle J_5^\mu(x) J_5^\nu(y) \rangle \sim \frac{C_J}{|x - y|^{2d-2}} \quad (2)$$

is not. Paper 14 Theorem obs is the statement that (??) cannot equal ball relative entropy through second order for generic axial sources. Dictionary improvement at linear order [?] does not install a bulk kinetic term.

Therefore a holographic UV that keeps a CFT axial current must contain a propagating bulk field dual to that current [?]. That field is not the Einstein–Cartan connection. Einstein–Cartan is an infrared limit, or it is the wrong theory for the axial channel.

2 The unique quadratic deformation

Let S_μ be an axial vector (the axial torsion dual, or an independent Proca field: the algebra below does not care). The most general local quadratic action with at most two derivatives, a mass term, and a linear Dirac source is, in momentum space,

$$\mathcal{L}_k = A \left(1 + \frac{k^2}{M^2} \right) |S(k)|^2 + B S(-k) \cdot J_5(k). \quad (3)$$

A has the dimension of a stiffness; M is the new mass; B is the Dirac–axial coupling. Completing the square,

$$S(k) = -\frac{B}{2A(1 + k^2/M^2)} J_5(k), \quad \mathcal{L}_{\text{eff}}(k) = -\frac{B^2}{4A(1 + k^2/M^2)} J_5 \cdot J_5. \quad (4)$$

The infrared lock is the M9.1 theorem [?]:

$$\frac{B^2}{4A} = \frac{3\kappa}{16}. \quad (5)$$

Then, and only then,

$$r(k) := \frac{\mathcal{L}_{\text{eff}}(k)}{-\kappa J_5 \cdot J_5} = \frac{3/16}{1 + k^2/M^2}. \quad (6)$$

Without (??) the infrared coefficient is a free parameter and M9.1 is lost. With it, the only remaining freedom in this ansatz is the mass M .

Theorem 2.1 (IR recovery). *Under (??) and (??), $\lim_{M \rightarrow \infty} r(k) = 3/16$ at every fixed k , and $r(0) = 3/16$ at every finite M . [P] (algebra).*

Theorem 2.2 (evasion of the contact obstruction). *At finite M , $r(k)$ is not constant. The position-space kernel is the Euclidean four-dimensional Yukawa function*

$$G(x) = \int \frac{d^4k}{(2\pi)^4} \frac{e^{ik \cdot x}}{k^2 + M^2} = \frac{M}{4\pi^2 r} K_1(Mr), \quad r = |x|, \quad (7)$$

which satisfies $(-\square + M^2)G = \delta^{(4)}$ and is not $\delta^{(4)}$ itself. [P] (Fourier analysis).

Theorem 2.3 (massless degeneration). *For any fixed $k \neq 0$, $\lim_{M \rightarrow 0} r(k) = 0$. A massless axial gauge field does not produce the Hehl–Datta contact. [P].*

Theorem ?? is the holographic tension made sharp. A *conserved* CFT current is dual to a *massless* bulk gauge field [?]. That theory is Einstein–Maxwell (axial), not Einstein–Cartan, and it has no contact infrared. A heavy field ($M \sim M_{\text{Pl}}$) matches (??) in the infrared and matches a *non-conserved* or high-dimension operator in the ultraviolet, not a conserved current at every scale. The obstruction is evaded only by changing the theory, exactly as Paper 14 said.

3 What M is allowed to be

If M is a few inverse fermis, spacetime torsion waves are laboratory fields. That is a different Lagrangian and is already excluded as Einstein–Cartan [?, ?]. The deformation is compatible with the program only if

$$M \gtrsim M_{\text{Pl}}, \quad (8)$$

so that torsion propagation is a Planckian correction and (??) is the effective theory below that scale. This is a consistency requirement, not a derivation of M . Ghost-free quadratic Poincaré-gauge kinetic terms exist in a restricted parameter set [?]; this note does not re-derive that classification and does not claim that every quadratic torsion kinetic term is healthy.

4 Machine check

Pre-registered gates C1–C5 were locked before the scripts ran. Solver: `m9_4_uv_axial.py`. Auditor (no solver import, sympy completion plus Bessel identities): `m9_4_audit_uv.py`. Both report PASS on C1–C5. The auditor confirms that without the infrared lock (??) the coefficient is not 3/16.

This is tree-level effective-field-theory algebra. It is not a loop computation and it is not a holographic Ward-identity calculation.

5 What this does not do

- It does not select $SU(3)_c \times SU(2)_L \times U(1)_Y$, three generations, Yukawas, or Λ . Those remain empirical input [?].
- It does not renormalize the metric sector. Einstein–Hilbert is still non-renormalizable. Quadratic curvature / Stelle / asymptotic safety are other Lagrangians, not derived here.
- It does not name a CFT. The AdS dictionary is cited [?], not constructed.
- It does not repair de Sitter (Paper IX).

- It does not make spacetime Hehl–Datta a laboratory field.
- It does not move a `MODELS.md` cell.

Admission. I did not invent a UV completion of the New Standard Model. I constructed the unique quadratic axial deformation that preserves M9.1 and evades the contact obstruction, and I checked the tree-level algebra two ways. Q4a remains [O]. Calling Q4b “a UV that works” is allowed only as infrared matching plus nonlocality, never as a selected microscopic theory.

6 Verdict

Claim	Status
Q4a selected pair / SM / metric UV	still [O]; not invented
Quadratic axial deformation (??)–(??)	derived
Infrared lock recovers 3/16	[P], audited
Yukawa kernel, not a contact	[P], audited
Massless limit is not Hehl–Datta	[P]
Identification $S_\mu =$ axial torsion	[C] (physical reading)
Final Status “UV completion” as a selected theory	unchanged, [O]

The phrase that is true: *the infrared of the unique quadratic axial deformation is the New Standard Model axial sector, and the ultraviolet of that deformation is not Einstein–Cartan.* The phrase that is false: *we now have the UV completion.*

References

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